Personal Project: NonTox

Criterion A: Planning

Strand i. State a learning goal for the project and explain how a personal interest led to that goal

"And They Lived Happily Ever After", this ten-syllable phrase was one that I would patiently await whenever I listened to my mother's bedtime stories. Bedtime stories and mythological epics were the first of my many childhood fascinations, and over time, I was able to develop this interest, eventually finding contentment in exploring a versatile range of concepts, ideas, and phenomena, through a variety of media, such as articles, newspapers, books, conversations, videos, and movies.

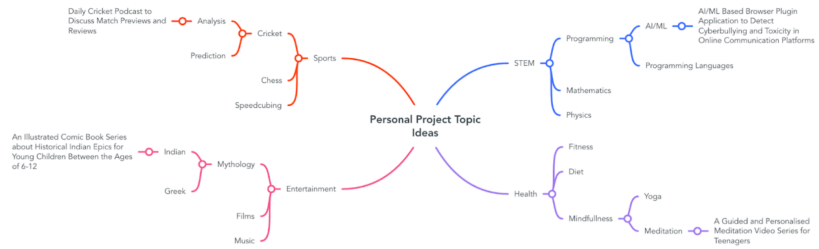


Figure A.1: Mind Map connecting my personal interests to potential learning and product outcomes.

Through this process, I had been able to discover my love for critical-thinking and problem solving, which was applied consistently in participating and securing top accolades in advanced level Mathematics Olympiads along with dabbling in Python Programming. During the past couple of years, with knowledge becoming increasingly accessible through advancements in search technology and the emergence of large language models, one of the most rewarding ideas I was able to explore was Artificial Intelligence (AI) and Machine Learning (ML). I read countless articles, watched awe-striking videos, and heard uplifting stories about the implementation of the power of Artificial Intelligence and Machine Learning, which is the broad project topic that I chose to focus on.

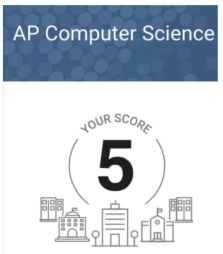


Figure A.2: Full Score on College CS Course

As mentioned before, I always had a deep interest in implementing critical thinking, creative thinking, and problem solving skills, whether it was solving complex algebra problems, coding certain operations in Python, or brainstorming general solutions to exciting riddles and conundrums. I was able to exercise this passion when I attempted a college-level computer science exam, scoring a full score of 5 out of 5 (Figure A.2). This dedication was only expanded when I started demonstrating my **Research skills**, by exploring AI and ML through multiple forms of media. My motivation to develop a concrete personal interest for this topic was sourced from the sheer connection that the aspects of this technology had with my engrossment in applying my critical thinking, creative thinking, and problem solving skills. This included the elements of essentially teaching computers 'how to think', actively deliberating upon choices and parameters during the implementation of these models, the opportunities to tackle potential bugs and learn from them, and their potential to be applied to solve real-world societal problems.

Furthermore, my main personal interest for exploring and learning more about the topic of AI and ML arose from a specific instigating event around a year ago. This was during a science and tech exhibition, during which I wanted to implement my newfound discovery of Machine Learning into resolving a pressing social issue, and I chose cyberbullying. As a part of the competition, due to time constraints as well as a lack of quality knowledge at the time, I only developed an elementary, non-coded model to make simple strides toward solving this societal issue. While I was unable to secure a winning result in the competition, I did have the opportunity to interact with an expert in the domains of AI and ML, who gave me a great deal of feedback as well as tips for better learning. Following the incident, when I was introduced to the Personal Project, I had a strong sense of attraction to pursue the areas of improvement from the tech exhibition, and ultimately explore the skills required to develop more complex and exhaustive models, completely reforming my initial idea in the process. This resulted in my learning goal for this project being, **to develop skills in**

programming Artificial Intelligence, specifically Machine Learning and Natural Language Processing Models, for Societal Benefit.

In the pursuit of this learning goal, I had already equipped myself with prior knowledge and skills in these domains over the past few years. Through Digital Design Classes in my Middle Years Program education as well as my own pursuit of knowledge, I had acquired a good sense of programming various sorts of applications in different languages, along with substantial experience with the Python programming language, as well as in HTML and CSS for web-development, which served as a base for further education towards the development of AI and ML Models. Furthermore, I was able to develop an immense sense of awareness toward the various societal issues surrounding us, such as environmental issues, rare diseases, as well as mental health, all motivating me to explore this learning goal and employ these technologies to solve some of the most pressing problems.

Specific (S)	• What: I want to learn how to develop models using AI, ML, and NLP, specifically for the purpose of using them for societal progress and benefit. • Why: I have seen from various research as well observations in the past that AI will allow us to solve many problems using its groundbreaking functionality. • Who: The main parties involved in my learning process will be myself, who will be consuming the content, and my supervisor, who will be providing me with expertise. • Where: The goal will mainly be achieved online due to the fact that these models are trained, tested, and validated using computers. • When: The goal will be achieved in a six month period from the month of June to the month of November.
Measureable (M)	I will measure my outcomes by: • Obtaining an evaluation from a Digital Design Facilitator. • Reflecting on my knowledge on the theory of AI models through various media. • Being able to explain and understand the code of the final AI/ML/NLP model.
Achievable (A)	• There are many more available resources and opportunities in AI/ML/NLP model development due to the rising prominence of these concepts, allowing me to gain large amounts of knowledge. • By putting in regular work along with appropriate and detailed planning, as well as support from my supervisor, I will have resources and motivation to achieve success.
Relevant (R)	• Programming AI and ML models has become incredibly influential and revolutionary as they can be applied in a plethora of domains. • There are various societal problems which lack concrete as well as major solutions that drive innovation.
Time Bound (T)	• Time Frame from June 2024 to November 2024 • Deadline of November 30th 2024 • Short-Term and Long-Term Tasks and Objectives
Challenging (C)	• I will have to manage and accurately follow a long and comprehensive schedule, featuring various tasks and objectives. • I have never programmed AI/ML/NLP models before, and they require a large quantity of code to be written.

Figure A.3: SMART-C Analysis of Learning Goal.

In order to further pursue this learning goal, my aim involved taking leverage of various resources and materials outside of my academic subjects and indulging in course material in various domains such as Introduction to AI, Applications of AI in Medical Industry, which was mainly associated with the societal impacts, Machine Learning, and Natural Language Processing for Sentiment Analysis. This would give me the ability to fulfill the parameters of my learning goal and apply my learnings to complete my product. Through the aforementioned process of procuring relevant knowledge, my plan involved being more aware of the applications of AI and ML in a vast array of areas and problems. This also aligned with the Global Context of **Scientific and Technical Innovation**, exploring **the impact of scientific and technological advances on communities and environments**, allowing to interpret the profound applicability of this learning process, especially in real-world scenarios, requiring critical solutions, and also acting as a way for me to exhibit by individual **Service as Action**.

Strand ii. State an intended product and develop appropriate success criteria for the product

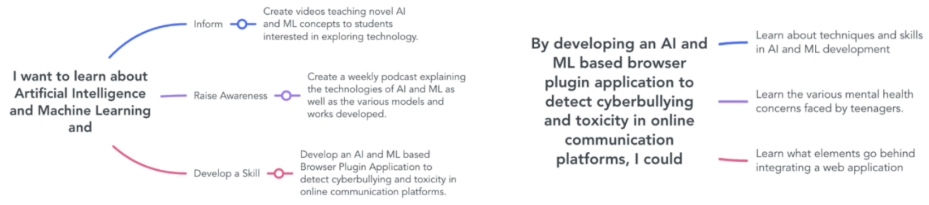


Figure A.4: Mind Maps Connecting Learning Goals to Possible Product Goals and Outcomes

In order to be able to pursue my learning goal to the greatest extent possible, I believed that it was important for me to make sure my product goal went hand in hand with it. That is why, I wanted to ensure that my final product involved the implementation of Artificial Intelligence, Machine Learning, and Natural Language Processing Models towards societal benefit.

In my mind, my product and its developmental process needed to involve hands-on experimentation as well as implementation of the relevant concepts, if knowledge and learnings had to be truly acquired. While, my product could have been about creating a series of videos about the implementation of AI, ML, and Natural Language Processing (NLP) models in different domains and industries, it would have lacked a direct approach, disabling me from understanding the profound theoretical concepts behind these models, as well as not giving me the opportunity to apply and implement these learnings in an area of choice.

I had identified one specific societal issue which Natural Language Processing models, models which are specialized in handling text data, could help mitigate, and it was cyberbullying.

Specific (S)	• What: I want to develop an AI/ML/NLP based browser plugin application to detect cyberbullying in online communication platforms. • Why: Cyberbullying is a problem which has had an adverse effect on numerous children, requiring the use of such unique solutions to offer a different approach. • Who: The main target audience which my product will be catered to is children aged 7 to 12, secondary stakeholders also include parents of children of these ages. • Where: The plan of the final product is to ensure that it will be accessible online in web stores, and mainly used in households and educational institutions. • When: The goal will be achieved in a six month period from the month of June to the month of November.
Measureable (M)	I will measure my outcomes by: • Interviewing and sending a survey to my target audience. • Creating, maintaining, and updating a cost accounting sheet. • Assessing the performance of the final AI/ML/NLP model.
Achievable (A)	• There are various types of browser plugin applications available, along with many courses, tutorials, and guides on developing them, from various sources, allowing me to create a highly-functional product. • There is a huge online Artificial Intelligence and Machine Learning community, along with various discussion forums, which will help me create a great model to perform its function, as well as fix any potential bugs.
Relevant (R)	• With children aged 7 to 12 being increasingly exposed to the internet and the cyberworld, issues associated with mental health and cyberbullying have risen. • This brings up the requirement of innovative solutions which haven't been implemented before, such as the very use of AI/ML/NLP models.
Time Bound (T)	• Time Frame from June 2024 to November 2024 • Deadline of November 30th 2024 • Short-Term and Long-Term Tasks and Objectives
Challenging (C)	• While I have worked on programming mobile applications and websites previously, I have never touched the development of browser plugin applications, which will result in taking significant time and effort to explore the dynamics of such programs in order to ensure successful completion. • I have never programmed AI/ML/NLP models before, and they require a large quantity of code to be written.

Figure A.5: SMART-C Analysis of Product Goal.

Cyberbullying, toxicity, and negative sentiment were all testing problems that I identified as highly prevalent in online communication platforms, in a verbal manner, as well as in a written one in the form of chat windows. As a result, my intended product goal for my personal project was **Developing an AI, ML, and NLP Based Browser Plugin Application to Detect Cyberbullying and Toxicity in Online Communication Platforms, for a target audience of Children aged 7 to 12.** In order to determine the success of my final product as well as to ensure and adjudicate whether the whole process of developing it would have been rewarding, a success criteria was going to be incredibly resourceful for the evaluation. My initial draft for the success criteria (Figure A.6), involved eight components, which worked on assessing various aspects of the product and the developmental process, allowing to evaluate the effectiveness of the AI model, the developmental process, as well as the working of the final browser plugin application. The highest priority from the success criteria was the aspect of Model Performance Metrics.

Criterion	Priority
Model Performance Metrics	High
User Interface	Low
User Experience	Medium
Function	High
Privacy and Safety	High
Customer	Medium
Aesthetics	Low
Cost	Medium

Figure A.6: Initial Success Criteria Draft

Criterion	1-2	3-4	5-6	7-8
Model Performance Metrics	The final testing metrics of the model performance for the parameters of accuracy, precision, F1, and recall achieve a range between 20% to 40%.	The final testing metrics of the model performance for the parameters of accuracy, precision, F1, and recall achieve a range between 40% to 60%.	The final testing metrics of the model performance for the parameters of accuracy, precision, F1, and recall achieve a range between 60% to 80%.	The final testing metrics of the model performance for the parameters of accuracy, precision, F1, and recall achieve a range between 80% to 100%.
User Experience	The overall experience of the application is underwhelming for the users, after operating the application and testing it on their unique preferences and use cases.	The overall experience of the application is mediocre for the users, after operating the application and testing it on their unique preferences and use cases.	The overall experience of the application is satisfactory for the users, after operating the application and testing it on their unique preferences and use cases.	The overall experience of the application is exceptional for the users, after operating the application and testing it on their unique preferences and use cases.
Cost	The total monetary cost incurred in the process of pursuing the learning and developing the product is greater than INR 4000.	The total monetary cost incurred in the process of pursuing the learning and developing the product is in a range between INR 2750 and INR 3000.	The total monetary cost incurred in the process of pursuing the learning and developing the product is in a range between INR 1750 and INR 2000.	The total monetary cost incurred in the process of pursuing the learning and developing the product is less than INR 1000.
Function	The final AI and ML based browser plugin application is ineffective in detecting cyberbullying and toxicity in online communication platforms.	The final AI and ML based browser plugin application is moderately effective in detecting cyberbullying and toxicity in online communication platforms.	The final AI and ML based browser plugin application is adequately effective in detecting cyberbullying and toxicity in online communication platforms.	The final AI and ML based browser plugin application is highly effective in detecting cyberbullying and toxicity in online communication platforms.
Privacy and Safety	The final AI and ML based browser plugin application is ineffective in presenting data privacy policies and protection information to the user group.	The final AI and ML based browser plugin application is moderately effective in presenting data privacy policies and protection information to the user group.	The final AI and ML based browser plugin application is adequately effective in presenting data privacy policies and protection information to the user group.	The final AI and ML based browser plugin application is highly effective in presenting data privacy policies and protection information to the user group.
Customer	The final AI and ML based browser plugin application is inaccessible to the target audience consumer group, which comprises children aged 7-12, who are dissatisfied with the product.	The final AI and ML based browser plugin application is moderately accessible to the target audience consumer group, which comprises children aged 7-12, who are moderately satisfied with the product.	The final AI and ML based browser plugin application is adequately accessible to the target audience consumer group, which comprises children aged 7-12, who are dissatisfied with the product.	The final AI and ML based browser plugin application is highly accessible to the target audience consumer group, which comprises children aged 7-12, who are highly satisfied with the product.

Figure A.7: Comprehensive Success Criteria

This included metrics such as Accuracy, Precision, Recall, and F1 Score,, all of which measure the performance of the AI model based on different factors. The other aspects which was of high priority were Privacy and Safety and Function, with the former being imperative towards ensuring that the data of users in online communication platforms, that was accessed by the product, wasn't compromised and stayed confidential, and the latter was also of paramount importance, as the end goal of the product was to develop a working browser plugin application integrated with an AI, ML, and NLP model, and it functioning according to the needs of the target audience was essential. Overall, this involved a trade-off between Form and Function, and I believed that having a browser plugin application integrated with an AI model which can complete the task of detecting cyberbullying in online communication platforms effectively, would have been much more impactful than a colourful interface featuring intricate styling choices, font families, colours, and aesthetics, that may not have been able to detect cyberbullying to a great potential, which resulted in me prioritizing Function over Aesthetics in the draft success criteria. Furthermore, there was also a trade-off between the User Experience (UX) and the User Interface (UI). The components of UX involve accessibility, navigation, usability, and structure, while UI involves interactivity, appearance, design, and colour theory. In line with my philosophy of Form over Function, I had decided to prioritize the User Experience over the User Interface, as my main requirement was to create a product that fulfilled its purpose of detecting cyberbullying in online communication platforms, rather than a product that looked pleasing to the eye.

Aspect	Justification + Research Derivation
Model Performance Metrics	Various onlines sources and articles have spoken volumes about the importance of model performance metrics, one in particular says, "By utilizing these metrics, we can optimize and fine-tune our models, leading to more accurate predictions and better decision-making processes.", which illustrates the importance of utilizing these metrics to improve and optimize the models. Overall, these metrics will allow to understand and evaluate whether the final AI/ML/NLP model is working to the standards it is expected to, and ensure that the model is effective in detecting cyberbullying and toxicity in online communication platforms.
User Experience	"User experience (UX) design is more than visual design — it's about making products and marketing materials accessible, and aims to offer an enjoyable, seamless experience.", these words are from Google, an innovative and multinational technology company, which illustrate the critical role the UX will have in the success of the product. By ensuring that the product provides all the elements of a proper UX, such as, a seamless and enjoyable experience, usability, accessibility, and structure, the users will be much more satisfied with its outcome and will not have any complaints with it. Furthermore, the experience that my product creates for the user will ultimately be vital to ensure that it does indeed detect cyberbullying and toxicity in online communication platforms, in a manner that is completely seamless and helpful to the user, so that it is able to achieve its main purpose and create a strong impact on them.
Cost	Ensuring that the costs incurred, during the entire process of acquiring necessary knowledge in the pursuit of developing the product, and the development of the product, stay within a strong and fixed budget is of paramount importance. This is due to the fact that I have limited savings, which have been collected over time in forms such as pocket money, and making sure that I wouldn't be having to incur any extra costs will not only allow me to complete the process in a fast and seamless manner, but will also allow me to work on aspects such as accounting and budgeting, to improve my self-management skills. Furthermore, as this product is mainly for the purpose of societal benefit, incurring lesser costs and exploring other ways to pursue my learnings and develop my product, in order to save imperative funds which can then be repurposed towards societal benefit and development.
Function	The function of the product is all about ensuring the product works the way it is expected to. In my case, making sure that my final AI and ML based browser plugin application does detect cyberbullying and toxicity in online communication platforms for children aged 7 to 12, both effectively and accurately, will allow me to evaluate whether my model is functioning and working the way it is supposed to. At the end of the day, the final goal of my product has to be to work as it is expected to, and that is by detecting cyberbullying and toxicity in online communication platforms, in order to fulfill its purpose and function as a unit and a technological system. While there may be many technical bugs and issues in the way of achieving a functional product, using my critical-thinking skills to overcome them, and obtain a final product that functions, by working the way I expect it to, is of paramount importance, to ensure the whole process of procuring learnings and developing the product has a meaningful and measurable outcome.
Privacy and Safety	For products such as this, there will always be a high reliance on the data of the users. In this case, the main user data that will be collected involves the messages and conversations transmitted over online communication platforms, as they will be sent through the main AI/ML/NLP model for detection of cyberbullying and toxicity in them. That is why, informing the users about what kind of data of theirs is being used and accessed by the application as well as taking consent of parents to access the data of their children is imperative, as failing to do so can result in lawsuits and fines imposed by governments and users, although only in extreme extents, as well as the application obtaining a bad reputation and losing the trust of the users. This makes it vital for the application to present all the privacy policy and data protection information to the users, as well as impose all required security features, in order to ensure the data of the users is not compromised, and guarantee that no legal issues and fines will be encountered, towards the overall success of the product and its features.
Customer	No product will be successful if it doesn't satisfy the needs of its customer and target audience. The main purpose of developing an AI and ML based browser to detect cyberbullying toxicity in online communication platform for children aged 7 to 12, is to make sure that these children aged 7 to 12 will actually be able to protect themselves and take action against cyberbullying, and it is important that these individuals will be satisfied with the product and find it accessible.

Figure A.8: Success Criteria Justification Derived From Research

These choices as well as prioritizations that I made all eventually allowed me to present my final success criteria (Figure A.7), to evaluate my product. I justified all six aspects of my success criteria (Figure A.8), which were Model Performance Metrics, User Experience (UX), Cost, Function, Privacy and Safety, and Customer. Through my justification, I was also able to showcase the specific, realistic, and rigorous nature of these aspects, and how they would guide me to evaluating the overall success of the product. Additionally, through the process of justifying each aspect of my success criteria, I was also able to actively demonstrate my **research skills**, found in the forms of both ***Information literacy*** and ***Media literacy***, through primary and secondary sources, to ultimately obtain a strong and well-rounded justification. Furthermore, I came up with qualitative and quantitative testing methods to evaluate each aspect of my success criteria (Figure A.9), in order to ultimately assess my product, its developmental process, and the procurement of learnings.

Aspect	Testing Method
Model Performance Metrics	Obtained and Visualized in the Programming Interface During Development
User Experience	Survey Sent to Target Audience of Children Aged 7 to 12
Cost	Maintaining and Updating Accounting Sheet
Function	Evaluation by Digital Design Facilitator
Privacy and Safety	Evaluation by Digital Design Facilitator
Customer	Survey Sent to Target Audience of Children Aged 7 to 12

Figure A.9: Testing Methods

Strand iii. Present a clear, detailed plan for achieving the product and its associated success criteria

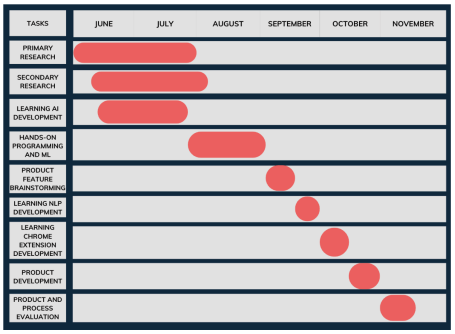


Figure A.10: Gantt Chart

With my personal project involving robust commitments to procuring my learnings and developing my product, with the whole process spanning multiple months, I knew for a fact that effective **self-management skills** and time-management were required to outline a clear set of deadlines and tasks, in order to achieve success and avoid procrastination, to be able to ensure that the project requirements would be completed on time.

To ensure that my entire process of procuring my learnings and developing the product for the personal project was seamless, I developed a series of long-term tasks to guide me with an essential comprehensive plan of all the objectives I needed to achieve effectively and productively during the process. I was able to visualize this series of long-term tasks in the form of a Gantt Chart (Figure A.10), which gave me an aesthetic, interpretable, as well as motivating view on the tasks that I needed to accomplish, allowing me to understand and recollect all my required objectives for the successful completion of the project, along with the required deadlines, and giving me an extra sense of motivation to finish these tasks with high-quality outputs, on time. By developing a comprehensive list of tasks as well as a Gantt Chart to visualize them, I was also able to gain immense clarity on the requirements as well as possible time-frames for further short-term tasks and objectives.

I had strategically planned all of my long-term tasks, keeping in mind that they had to go simultaneously with other commitments that I had expected, such as vacations and school assessments. That is why, in the months of June and July, which majorly consisted of my school's summer vacations, and a minute amount of school coursework in the initial stages of the new academic year, I focused on my primary and secondary research. These were tasks which I had significant prior experience with as a part of my prior MYP studies, and for this reason were suitable for this specific time-frame, during which I was able to seamlessly complete the tasks, while **balancing** them with various travel and family activities. I was able to commit a lot of my time to learning AI development, making use of summer courses and programs in these endeavours, as well as continuing and completing my primary and secondary research tasks to gain much more informed insights on the entire problem area. This continued in the month of August, where I committed large amounts of my time to hands-on programming and machine learning, which involved me actively implementing my past learnings with real-time and impactful projects. During the month of September, I had great amounts of school coursework, assignments, and assessments, which disabled me from contributing a lot of time for my personal project, leading me to pursue a less enduring task, in the form of brainstorming features for my product, which allowed me to make progress on both my personal project as well as my academic commitments at the same time. Following the conclusion of my exams, throughout the month of October and in the early part of the month of November, I invested a majority of my time working on actively developing the product. The main reason for this was because my school's Personal Project Exhibition was scheduled during the first few weeks of November, which gave me a sense of motivation to complete my product and have it ready to present it to my target audience and stakeholders on the day of the Exhibition.

Following the planning of my long-term tasks, I was able to further divide these long-term tasks and their allotted timeframes into an exhaustive list of short-term objectives (Figure A.11). This was along with information on the estimated time for completion as well as the relevant success criteria aspects, which further gave me a sense of understanding of my workload, to plan my schedules accordingly, and motivation to complete and achieve all the required objectives for my personal project in a successful manner, also allowing me to exhibit my **Organisation skills**. Due to my extensive academic commitments as a part of the MYP program as well as various extracurricular activities that I had undertaken in, effectively planning out all my available time for the personal project ended up allowing me to me much more productive, as well as obtain a deeper understanding on the specific requirements associated with my project, as well as my overall progress over time.

Long Term Objective	Short Term Objective	Relevant Success Criteria	Estimated Completion Time
Primary Research	Send Survey to Parents of Children Aged 7 to 12 on Cyberbullying	Function	3 Days
	Interview with Children Aged 7 to 12 on Preferred Solutions to Tackle Cyberbullying	Customer	2 Days
	Focus Group with Grade 10 Peers Reflecting on Past Experiences of Cyberbullying and Discussing Potential Solutions	Function	1 Day
	Interview with Machine Learning Expert to Explore Detecting Cyberbullying and Toxicity in Online Communication Platforms	Model Performance Metrics Function	1 Day
Secondary Research	Read 5 Media Articles on the Applications of AI for Societal Benefit	Function Customer	2 Days
	Read 3 Research Papers on the Implementation of Specific AI Models in the Domains of Healthcare and Mental Health	Function Customer	2 Days
	Watch 2 Videos Explaining the Theory Behind Specific AI Models and Their Applications for Societal Benefit	Function Customer	3 Days
Learning AI Development	Course on Introduction to AI	Model Performance Metrics Function Cost	2 Weeks
	Course on Medical AI to Explore Societal Benefit	Model Performance Metrics Function Cost	3 Weeks
Hands-On Programming and ML	Practice Existing Python Programming Skills	Model Performance Metrics Function	1 Week
	Program 3 Python Practice Projects	Function	1 Week
	Course on Machine Learning	Model Performance Metrics Function Cost	1 Week
	Program 1 Machine Learning Practice Project	Function	1 Week
Product Feature Brainstorming	Interview Digital Design Teacher on Possible Features for the Product	Function User Experience	2 Days
	Consult Supervisor on Possible Features for the Product	Function User Experience	2 Days
	Send Survey to Children Aged 7 to 12 on Possible Features for the Product	Customer Function User Experience	3 Days
	Research and Analyse Existing Solutions for Detecting Cyberbullying in Online Communication Platforms	Function User Experience	1 Week
Learning NLP Development	Course on NLP for Twitter Sentiment Analysis	Model Performance Metrics Function Cost	1 Week
	Program NLP Project for Sentiment Analysis of Online Tweets	Function	1 Week
Learning Chrome Extension Development	Course on Chrome Extension Development	Function Cost User Experience	1 Week
	Program Practice Chrome Extension of Hello World	Function	1 Week
Product Development	Program AI/ML/NLP Model to Detect Cyberbullying and Toxicity from Text Inputs	Function Model Performance Metrics Customer	3 Days
	Program Chrome Extension to Log Messages in Chat Windows of Online Communication Platforms	Function Customer User Experience	5 Days
	Integrate the AI/ML/NLP Model with the Chrome Extension to Achieve a Final Product	Model Performance Metrics Function Customer User Experience Cost	2 Days
	Identify and Implement All Required Privacy Policies, Protection Information, and Verification Requirements	Privacy and Safety Customer User Experience	2 Days
	Receive and Implement Relevant and Necessary Feedback from Personal Project Mentor	Model Performance Metrics Function Privacy and Safety Cost Customer User Experience	2 Days
Product and Process Evaluation	Collect and Visualize Metrics to Evaluate the Performance of the Model	Model Performance Metrics	2 Days
	Present and Interact with the Digital Design Facilitator to Obtain Evaluation and Feedback on the Function, and Privacy and Safety of the Product	Function Privacy and Safety	3 Days
	Finalize and Review Cost Accounting Sheet	Cost	3 Days
	Send Survey to Children Aged 7 to 12 to Evaluate the Product Based on the Success Criteria	Customer User Experience Privacy and Safety	3 Days
	Present Product and Process to Parents of Children Aged 7 to 12 at the School's Personal Project and Obtain Evaluation and Feedback	Model Performance Metrics Function Privacy and Safety Cost Customer User Experience	3 Days

Figure A.11: Long-Term and Short-Term Action Plan

Criterion B: Applying Skills

Strand i. explain how the ATL skill(s) was/were applied to help achieve their learning goal

A set of skills which contributed a significant amount to my learning process were **Research Skills**, and specifically, **Information literacy skills**. As a part of my process of Learning AI and NLP Development, I completed a course titled, 'NLP: Twitter Sentiment Analysis' (Figure B.1), wherein I was able to actively ***understand and use technology systems*** such as AI, ML, and NLP, exploring their theory as well as practical applications, to obtain invaluable learnings, taking imperative steps towards the successful achievement of my learning goal.

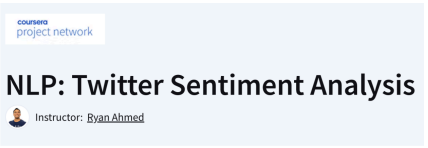


Figure B.1: Course on AI, ML, and NLP programming.

Additionally, as a part of my secondary research, in order to be able to further explore the various applications of AI technologies in performing a variety of tasks, mainly for societal benefit, I strived to ***collect, record and verify data***, mainly in the form of graphs (Figure B.2). These graphs explored the growing prominence of AI systems and technologies and startups over time, as well as a few of the most popular application areas of AI, such as Natural Language Processing and Voice Assistants. Through the process, I came across profound realizations, ultimately being able to ***access information to be informed*** on the sheer groundbreaking nature of AI technologies and the power of applying them to solve problems in a multitude of domains. Most importantly, I gained indispensable knowledge on the ways through which technologies such AI and ML could be applied to achieve the greater good of societal benefit.

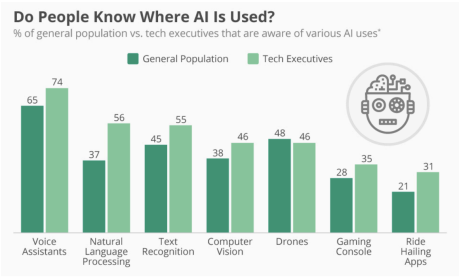


Figure B.2: Graph data collected on the various applications of AI.

Primary Sources	Secondary Sources
Interviews → Empathizing with peers on problems solved by the applications of AI/ML/NLP models. → Subject-Matter experts to learn about helpful resources and coding techniques for AI/ML/NLP model development. Surveys → Understanding societal problems which have the greatest requirement for innovative solutions. Research Papers → Applications of AI in the domain of Mental Health. → Evolution of Artificial Intelligence Technology over the years,	Articles → History and importance of Artificial Intelligence, Machine Learning, and Natural Language Processing. → The capabilities and impacts of AI/ML/NLP models for societal benefit. Onlines Courses → Natural Language Processing Development for developing the AI/ML/NLP model. → Medical AI Development to understand areas of implementation of AI for societal benefit. Videos → TED Talk on the promises and perils of AI. → YouTube Tutorial on Machine Learning Projects Development.

Figure B.3: Primary and Secondary Sources

After spending multiple years in my MYP students, I knew for a fact that identifying resources which would assist in my learning process as early as possible was important, which prompted me to ***identify primary and secondary sources***, (Figure B.3) that guided me in understanding the world of AI/ML/NLP models, their applications for societal benefit, and the problems which they had the ability to solve. The identification of these sources allowed me to structure as well as track my learning and research progress much more effectively, and ensured that I was able to successfully achieve my learning goal.

I was also able to exhibit **Media literacy skills** while achieving my learning goal. This was mainly done by ***demonstrating awareness of media interpretations of events and ideas***, which was implemented in the process of exploring various online articles, from different sources and news agencies across the world, which were about the use of artificial intelligence in a variety of fields as well as the main future and potential of AI. I read articles from Washington Post (Figure B.4), based in the United States of America, BBC (Figure B.5), based in the United Kingdom, and the Economic Times (Figure B.6), based in India, and explored their respective interpretations of the ideas of AI, ML, and NLP, to obtain diverse as well as versatile learnings on the various applications of such models for societal benefit, along with their future scope. I ended up ***locating, organizing, analysing, evaluating, synthesizing and ethically using information from a variety of sources and media***, and in the process, ***sought a range of perspectives from multiple and varied sources***.



Figure B.4: Washington Post article from USA on the AI revolution.



Figure B.5: BBC article from UK on the capabilities of AI.



Figure B.6: Economic Times article from India on the potential of AI.

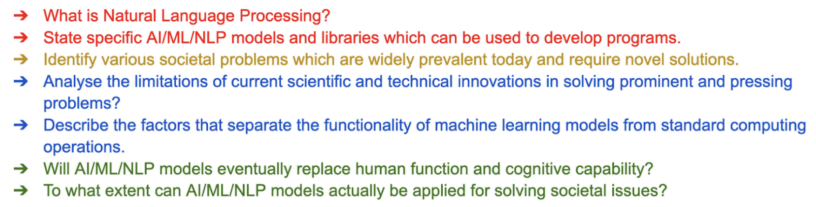


Figure B.7: Inquiry and Research Questions

Next, I also worked on actively implementing my critical thinking skills, by carefully ***formulating factual, topical, conceptual and debatable questions*** (Figure B.7), to assist with my inquiry and research. After formulating them, my aim involved finding answers to these questions through all the methods of research, including primary and secondary sources, which I was going to utilize. Some were answered after interactions with Subject-Matter experts and peers, while others were answered through the use of search engines, articles, and research papers, with all my findings aiding me in ***drawing reasonable conclusions and generalizations*** as well as making significant strides towards achieving my learning goal, which revolved around these topics and concepts.

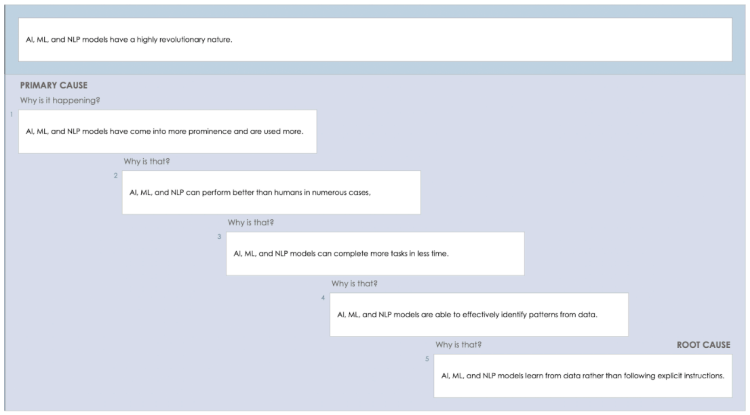


Fig B.8: 5-Why Analysis

Additionally, I explored my **Creative-thinking skills** by ***using brainstorming and visual diagrams to generate new ideas and inquiries***. I achieved this by conducting a 5-Why Analysis (Figure B.8), to learn about, understand, and reflect upon the groundbreaking and revolutionary nature of AI, ML, and NLP models, and came to the conclusion that the root cause or contributing factor to this was the models learning from data rather than following explicit instructions.

I also demonstrated **Transfer skills** by creating a mind map (Figure B.9), which explored how AI, ML, and NLP development was an intersection of different areas of study, including Mathematics, Computer Science, and Physics, allowing me to learn and reflect about the theoretical concepts associated with the programming of such models and ***make connections between subject groups and disciplines***, in the process.

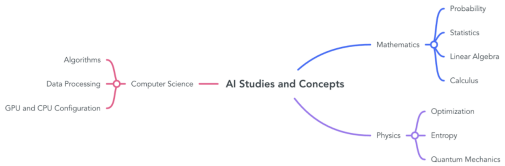


Figure B.9: Mind map connecting concepts of CS, Math, and Physics behind AI.

During my process of procuring knowledge to achieve my defined learning goal, my application of **Communication skills** was incredibly resourceful, as I exercised them in a variety of scenarios. I decided to being my procurement of learnings by exploring the various applications of Artificial Intelligence and Machine Learning Models for Societal Benefit, allowing it to serve as a foundation for my objective of Learning AI Development (Figure A.10), giving me a conscious reminder of the concrete real-world impacts the implementation of these models would have on the society around me. In this process, I pushed myself to ***read a variety of sources for information and for pleasure***, with the most notable one being a research paper, published in the Frontiers Journal (Figure B.10), which explored the application of AI for the improvement of mental health, a highly pressing social issue, associated with problems such as cyberbullying and depression, allowing me to gain great amounts of knowledge in this topic, which I didn't explore earlier.



Figure B.10: Research Paper on the role of AI in improving Mental Health.

In order to better represent my learnings as a culmination of this process of research and learning, I **took effective notes**, to ultimately ***make inferences and drew conclusions***, in the form of a mind map (Figure B.11), on the various applications of Artificial Intelligence for Societal Benefit, in all sorts of domains such as healthcare, climate, and education, which gave a me a profound sense of motivation to implement the technologies in such areas, and allowed me to reflect upon my learnings.

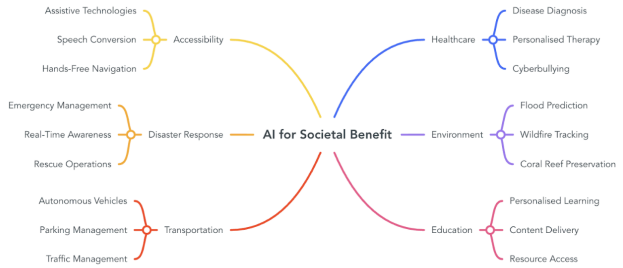


Figure B.11: Mind map collating notes on the applications of AI for Societal Benefit.

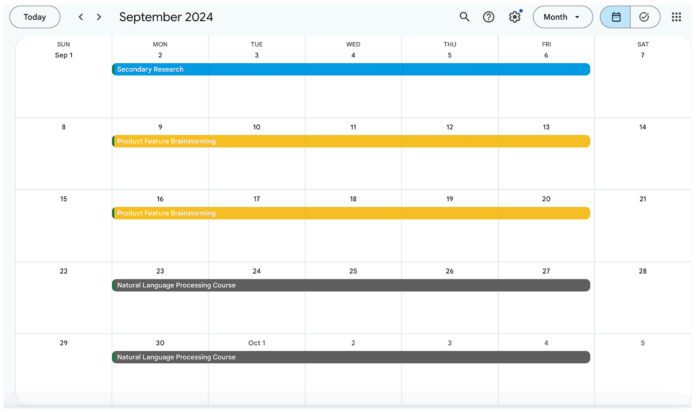


Figure B.12: Calendar for maintaining tasks.

Self-management skills, and specifically **Organization skills**, were monumental in ensuring my learning goal's success. I ensured to ***plan short- and long-term assignments***, as well as ***keep and use a weekly planner* for assignments**, in the form of a calendar (Figure B.12), to obtain active updates on those tasks which I needed to do, as well as to maintain a sense of progress throughout the learning process, to ensure I adhered to my action plan (Figure A.11), ***ultimately planning strategies and taking action to achieve personal and academic goals***.

I exercised my ***Affective skills*** during the process, mainly by ***practising focus and concentration***. As I had a lot of academic tasks, personal commitments, and most importantly, course-work and programming practice for my personal project, managing all my objectives and ensuring I completed all of them productively, effectively, as well as punctually was imperative. That is why, I followed the Eisenhower Matrix (Figure B.13), a task management and productivity tool, allowing me to sort my tasks, based on their significance in the context of acquiring knowledge to achieve my learning goal, into four categories based on importance and urgency - Do, Delegate, Decide, Delete.

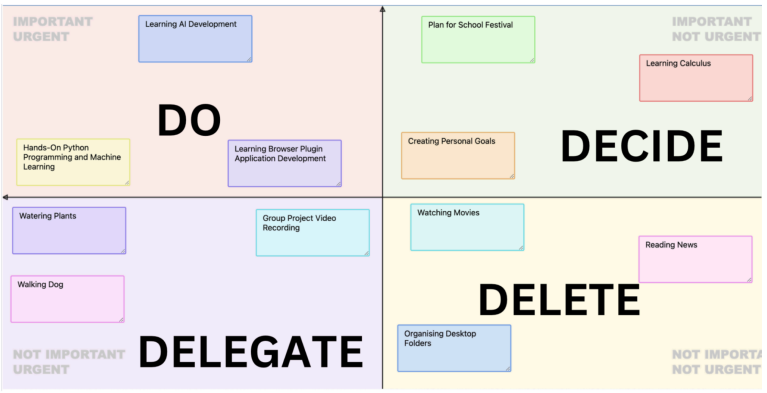


Figure B.13: Eisenhower Matrix to prioritize tasks.

Practicing **Reflection skills** ended up being incredibly resourceful for me during my process of working towards achieving success with my challenging learning goal. In order to learn the working and theory behind AI, ML, and NLP models, I decided to practice an unconventional learning method known as the Feynman Technique (Figure B.14). As a part of the technique, I started by noting down all knowledge on the topic, then tried explaining the concept to myself in my own simple words, which was followed by cross-referencing the information from my courses and past research, to ultimately understand the concepts very well, through a hands-on and experimentative manner , and as a result, I ***developed new skills, techniques and strategies for effective learning***.

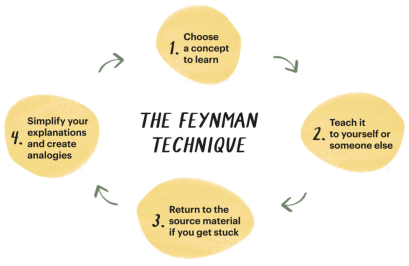


Figure B.14: Using and understanding the Feynman Technique.



Figure B.15: Empathizing with a Student

As a part of primary research, I developed my **Collaboration Skills**, wherein I interviewed a high school student (Figure B.15), who actively benefited from the use of novel Generative AI and Natural Language Processing technologies. Through our interaction, I came to know about the various problems they faced, involving a lack of productivity and difficulty acquiring knowledge in an understandable manner, and how they were solved through the implementation of these technologies and tools. Through this process I learnt about the main aspects, features, and model architectures associated with successful AI/ML/NLP models, and as a result, was also able to ***practise empathy***.

Finally, I also had the great opportunity to interview an IT manager (Figure B.16), working on software projects involving the implementation of AI technologies for the benefit of the society, specifically for the residents of the State of Colorado, in the United States of America. Through our interaction, I not only ***listened actively to other perspectives and ideas***, but also was able to get an in-depth technical insight on the specific types of AI, ML, and NLP models used for specific tasks, an understandable explanation on the theory behind them, as well as general tips to achieve success in the programming process, with all of these having an invaluable impact on my learning process.



Figure B.16: Interviewed an individual working on employing AI technologies for societal benefit.

<u>Strand ii. explain how the ATL skill(s) was/were applied to help achieve their product goal</u>

Next, I was able to pursue the development of my **Information literacy skills** in a plethora of manners, throughout the process of developing my product. As a part of my process for brainstorming features for my product as well as primary and secondary research to understand the problem of cyberbullying and its effect on children aged 7 to 12, I created and sent out a survey (Figure B.17 and Figure B.18), to parents of these children, and tried to get insights into the importance of the product and the demand of the user group and stakeholders, as well as the overall opinion parents regarding the notion of monitoring online activity, and as a result was able to ***process data and report results***. This process also allowed me to ***collect and analyse data to identify solutions and make informed decisions*** on the overall scope and importance, potential impact, and features of my product.

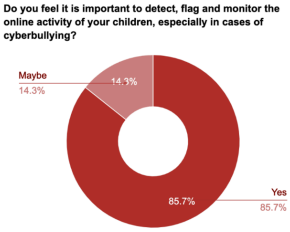


Figure B.18: Survey Results.

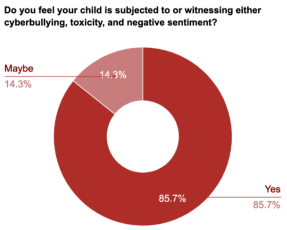


Figure B.17: Survey Results.

Citation	Source	Outcome	Success Criteria
AI. "Decoding Depression: How AI Is Revolutionizing Mental Health | Mariam Khayretdinova | TEDxBoston." *YouTube*, 2024, www.youtube.com/watch?v=32YgTYMhT4o.	Video	Applications of AI in Mental Health and Cyberbullying, and key mental health and abuse problems faced by individuals.	Consumer
Milosevic, Tijana, et al. "Artificial Intelligence to Address Cyberbullying, Harassment and Abuse: New Directions in the Midst of Complexity." *International Journal of Bullying Prevention*, vol. 4, no. 1, Springer Science and Business Media LLC, Feb. 2022, pp. 1–5, https://doi.org/10.1007/s42380-022-00117-x.	Research Paper	Current scope of AI/ML/NLP models in addressing cyberbullying and online abuse, and the challenges and significance associated with this task.	Function
"Free Chrome Extensions Tutorial - Free Chrome Extensions Development Tutorial." *Udemy*, 2024, www.udemy.com/course/chrome-extension-dev/.	Course	Developing Chrome Extensions, a type of browser plugin applications.	User Experience
Robinson, Alon. "Communicating between Webpages and Chrome Extensions." *Medium*, 22 Jan. 2016, medium.com/@AlonNola/communicating-between-webpages-and-chrome-extensions-f32c326bbfd9.	Article	Allowing browser plugin applications to access information from online communication platforms in webpages.	Function
"RichardRivaldo/Sentiment-Analysis: Sentiment Analysis Classifier with Machine Learning Models." *GitHub*, 2024, github.com/RichardRivaldo/Sentiment-Analysis.	Repository	Different AI/ML/NLP models and programming techniques used for sentiment analysis for detecting cyberbullying.	Model Performance Metrics

Figure B.19: Product development bibliography table.

Throughout the process of developing my product, I came across vast amounts of research papers, videos, articles, studies, repositories, and tutorials, which all gave me great insights as well as suggestions for the development of my final product, allowing me to get introduced to other applications, products, or services working to solve the same problem, as well as implementations of AI, ML, and NLP in similar domains, which were all applicable in this scenario. In line with the philosophy of Academic Honesty, I ***created references and citations, used footnotes/endnotes and constructed a bibliography according to recognized conventions***, citing my sources in the MLA 9 format (Figure B.19), and obtained a collection of all the sources that helped in the creation of my product, allowing me to refer to them at any time.

Source 1	
Browser Extension | Wikipedia	
Origin	**Purpose**
→ The source is an online article from Wikipedia, an online encyclopedia managed by a group of volunteers. → There is no information on the name of the author or publisher as well as the publishing date of this source.	→ To inform the readers about the definition, history, and challenges associated with browser extensions and plugin applications. → Offers an objective view into the functionality as well as use-cases of these technologies.
Value	**Limitations**
→ The source does help understand the basic functionality and implementations of browser extensions and plugin applications. → There is restricted value to the project as it doesn't provide information on code or technical resources which can assist in the development process.	→ Information in Wikipedia can be edited by anyone, hence this information might be biased or inaccurate. → The overall perspective is quite general and doesn't explore relevant topics or ideas.

Figure B.20: OPVL Analysis for Source 1

For the successful development of my product, I knew that I needed to traverse through the various techniques, forms, and implementations of browser plugin applications. To satisfy this requirement, I came across two different articles, and I wanted to ensure that I would be using the source which was more unbiased, content-rich, revered, and reputable, and for that reason, I ***evaluated and selected information sources and digital tools based on their appropriateness to specific tasks***, as well as, ***used critical-literacy skills to analyse and interpret media communications***, using an OPVL analysis (Figure B.20 and Figure B.21), which stands for origin, value, purpose, and limitation.

Source 2	
What Are Chrome Extensions? | GeeksforGeeks	
Origin	**Purpose**
→ The source is an online article from GeeksforGeeks, a specialized Computer Science Portal offering learning resources and articles. → The article was published on the 22nd of February, in the year 2024.	→ To inform the readers about the definition, function, and a few examples of browser extensions and plugin applications. → Providing sample code as well as further resources for the development of browser extensions and plugin applications.
Value	**Limitations**
→ The source does help understand the functionality as well as the specific code behind the development of these applications. → There is immense value to the project as it allows to set a framework for the development of an extension, which can be upscaled to the final product.	→ There isn't a lot of information on the identity and qualifications of the author, apart from their username. → Otherwise, the source isn't limited with regards to its reliability as it is used by IT professionals for job interviews and problem-solving.

Figure B.21: OPVL Analysis for Source 2

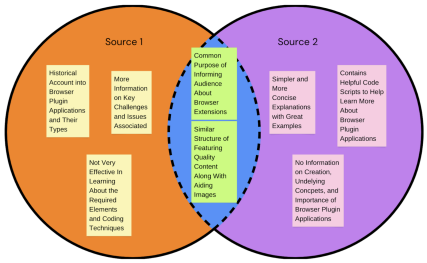


Figure B.22: Venn Diagram

In order to refine my analysis and represent it in a more visual manner, I also implemented my **Media literacy skills**, ***comparing, contrasting, and drawing connections among media resources***, by creating a Venn Diagram (Figure B.22), which allowed me to understand the individual strengths and weaknesses of both the sources, as well as the aspects which contributed towards their commonalities, to ultimately gain insights on the overall reliability of the sources, and the extents to which they were respectively effective. After both the analyses, I had decided to choose Source 2 over Source 1, as my guide to understanding and implementing browser plugin applications and extensions. This was mainly due to the fact that Source 1 was offering great potential for bias and inaccuracy, as well as an incredibly general and limited overview into the topic, while Source 2 was much more credible, content-rich, and helpful, overall offering incredible value to uncovering this engrossing topic.

I was also able to implement my **Thinking skills** in order to successfully accomplish my product goal. Specifically, I worked on the utilization of my **Critical-thinking skills**, wherein I preserved to ***troubleshoot systems and applications***, by ***identifying obstacles and challenges*** that were acting as a wall separating me from a successful product, and fixing key bugs which restricted my AI/ML/NLP model from functioning. While I faced plenty of such situations during the process of developing my model, one in particular came during the stage of preprocessing the datasets used for training the model, wherein I needed to plot a World Cloud to visualize my dataset, and faced challenges in doing so, mainly due to absence of a few vital libraries and variables. I ended up resolving this by referring to the documentation of these libraries and the error messages, to ultimately plot my final world cloud (Figure B.23).



Figure B.23: Transition from Code Error to Completed Word Cloud



Figure B.24: Deployed Browser Plugin Applications

Following that, the implementation of **Creative-thinking skills** proved extremely handy and were incredibly relevant to the overall idea and function of my product. The entire idea of my product as well as the notion of browser plugin applications were established from the concept of ***designing improvements to existing machines, media and technologies***, with its purpose of detecting cyberbullying, toxicity, and negative sentiment in the text transmitted over online communication platforms enhancing the capabilities and features of these existing technologies, as well as providing more functionalities in browsers (Figure B.24), along with other browser extensions and plugin applications. With the addition of browser plugin applications and extensions such as my product, the functionality of various online communication platforms was able to develop from not just a place to share ideas and information, but also a monitored and secure hub, possessing the ability to curb signs of cyberbullying, toxicity, and negative sentiment, for children aged 7 to 12.

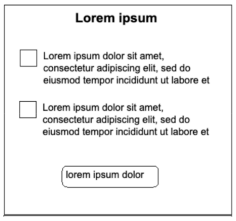


Figure B.25: Privacy Information Page Wireframe



Figure B.26: Message Logs Page Wireframe

In order to ensure the overall development of the above product was seamless, I understood the importance of ***using brainstorming and visual diagrams to generate new ideas and inquiries***, and ended up creating wireframes of the different pages I was going to include, beforehand. This helped understand the required elements for both the interface and the experience, and also assisted in obtaining a summary of the features I was going to include in the browser extensions, serving as a constant reminder during the developmental process. Ultimately, I designed two wireframes, one for the Privacy Information Page (Figure B.25), and the other for the Message Logs Page (Figure B.26), and carried out the entire process effortlessly due to my prior knowledge from MYP Digital Design classes, thereby being able to ***apply existing knowledge to generate new ideas, products or processes***.

Additionally, throughout the entire process of developing my product, I tried to ***inquire in different contexts to gain a different perspective*** on the entire conception of an AI/ML/MLP based browser plugin application to detect cyberbullying in online communication platforms. I demonstrated this by creating a mind map (Figure B.27), featuring different perspectives and explorations associated with inquiry in the areas of the six IB MYP Global Contexts, that I was able to successfully conduct and investigate as a part of this overall prolific product development journey.

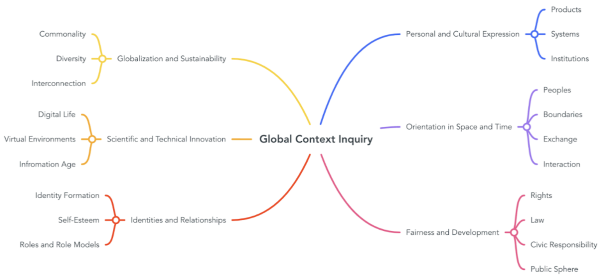


Figure B.27: Global Context Inquiry Mind Map

Next, **Self-management skills**, and specifically **Organization skills** were essential in ensuring I achieved my product goal. By ***setting goals that were challenging and realistic***, using a SMART-C Analysis (Figure A.9), I was not only inspired to accomplish my tasks according to my specifications, deadlines, and measurable testing methods, but was also able to identify various challenging aspects which I prioritized tackling. This, along with the realistic nature of it that I had identified, which gave me a sense of direction on the key aspects of my idea which strengthened it as well as helped me stay committed to achieving the goal, motivated me to a great extent, and made me ensure that I responsibly followed the schedule which I had set for myself, allowing me to accomplish all the tasks efficiently and appropriately.

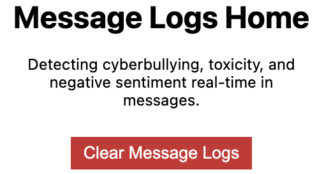


Figure B.28: Screenshot of Working Product.

With the entire idea and concept of my product, and the whole process of developing it, **Affective skills** were utilized extensively, and this was achieved mainly by ***practising strategies to prevent and eliminate bullying***, going hand-in-hand with my product (Figure B.28), an AI, ML, and NLP based browser to detect cyberbullying and toxicity in online communication platforms. With the main functionality of my product being the detection of potential cyberbullying, toxicity, and negative sentiment in the text transmitted over online communication platforms, and then flagging or reporting the case, I was able to create a mechanism that made significant steps towards mitigating the ever-threatening effects of bullying.

Collaboration skills also proved helpful while working on the successful achievement of my product goal. Throughout the process of developing my product, I ***listened actively to other perspectives and ideas***, by actively collecting feedback on the features of my browser plugin application from my peers in my Digital Design class (Figure B.29), mainly for the reason that they also possessed reasonable experience with the development of websites and applications, as well as the various content, features, and aesthetics associated with them. All of these inputs were incredibly resourceful, and helped improve the quality of my final product, against the success criteria.

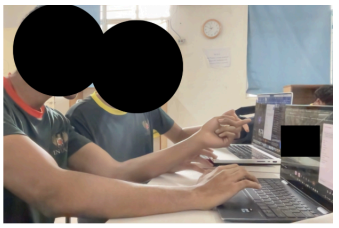


Figure B.29: Discussion with Peer

I knew that the final product that I was going to create and deploy needed to have features which were sourced from a great sense of understanding, importance, and observation. I was able to develop this sense of perception, by participating in a week-long Leadership Development Program (Figure B.30), wherein I encountered various tasks and challenges, often witnessing or engaging in minor disagreements or disputes, requiring me to ***manage and resolve conflict, and work collaboratively in teams***. Subsequently, I gained profound insights on proactive and effective methods of identifying, managing, and mitigating bullying and toxicity, which I was later able to inculcate into the features and functionality of my AI/ML/NLP model and browser plugin application, to obtain a product which was well-thought and deeply-rooted through active experimentation.



Figure B.30: Leadership Program Certificate

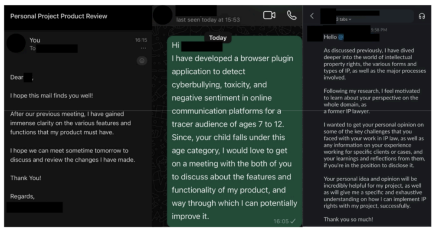


Figure B.31: Outlook

Figure B.32: Whatsapp

Figure B.33: Slack

Furthermore, throughout the process of developing and improving my product, I actively worked on outreach, by ***using social media networks appropriately to build and develop relationships***, such as Outlook (Figure B.31), to formally communicate with my Personal Project Supervisor, Whatsapp (Figure B.32), to communicate with a family member in an informal manner, and Slack (Figure B.33), to professionally communicate with an individual in a large organisation. As a result, I was not only able to enhance the features and functionality of my product, but also improved my association and bonds with these individuals.

During my process of developing my intended product, **Communication skills** came in handy, and boosted the process by a great extent. As a part of developing my AI/ML/NLP model to detect cyberbullying as the base for my browser plugin application, I needed to ***understand and use mathematical notation***, specifically the concept of Bayes' Theorem (Figure B.34), in the context of calculating the probability of the input text constituting to cyberbullying, toxicity, and negative sentiment.

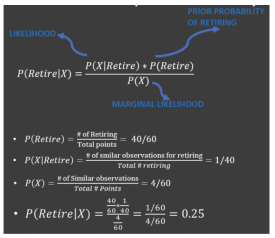


Figure B.34: Understanding and Implementing Bayes' Theorem.

Similarly, as a part of my efforts to pursue hands-on Machine Learning (ML) Development, I ***used and interpreted a range of discipline-specific terms and symbols***, including models and equations (Figure B.35), such as supervised learning, regression, classification, features, variables, and cost functions, setting a framework for the final model aspect of the product, and assisting with developing the code for my AI/ML/NLP model to detect cyberbullying, in a manner which I would be able to interpret.

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_{w,b}(x^{(i)}) - y^{(i)}\right)^2$$

Figure B.35: Symbols and Terms associated with Machine Learning.

Finally, by having frequent meetings with my Personal Project Supervisor and Mentor (Figure B.36), I ***received meaningful feedback*** on my product development process, code and documentation implemented, and most importantly, the features and user experience of my product, to help improve the final output, and subsequently achieve success according to the success criteria (Figure A.6).


Figure B.36: Feedback from mentor on the features of the product.

Criterion C: Reflecting

Strand i. Explain the impact of the project on themselves or their learning

Through the whole process of pursuing my learning and developing my product as a part of my personal project, I not only achieved both my learning and product goals, but also increased my confidence as well as skillset in my overall abilities in programming. I dove incredibly deep into the intricacies of my personal interest in the areas of Artificial Intelligence, Machine Learning, and Natural Language Processing, and became quite proficient in developing such models for a variety of tasks and applications. Moreover, following the personal project, I increased my awareness of and empathy towards a variety of societal problems which were prevalent across many demographics and scenarios, and the significance as well as real-world implications that they held. Most importantly, I had been able to more effectively develop a strong

sense of understanding on the topics of cyberbullying and online toxicity, being able to empathise with the problems faced by children aged 7-12. At the end, I was equipped with a stronger as well as more holistic resolve and motivation to develop innovative and novel solutions to solve society's most pressing problems.



Figure C.1: Exhibition Feedback

As a result, I became much more **Knowledgeable,** following the procedure of ***engaging with issues and ideas that have local and global significance***. Additionally, after sincerely empathizing with my peers, target audience, as well as many individuals across the globe to develop a novel and beneficial product, I transformed myself into a **Caring** individual, one who ***acts to make a positive difference in the lives of others and in the world around.*** This fact was able to sink in much better during my school's Personal Project Exhibition, wherein I was able to receive vital feedback and comments on my progress from a range of parents and students (Figure C.1), through which I was able to understand the significance of the problem I was tackling as well as the potential my solution had to create a profound positive societal difference.


Figure C.2: Presentation to MYP Year 2 Classroom

One of the biggest and most solid takeaways for me as a result of this personal project, was that when pursuing the development of a solution for a problem, especially of a scientific and technical innovation, everything needs to be centered around the audience. While this realization was sourced throughout the activities of the process of working on the project, I felt it was best exemplified during one particular experience, which was during my primary research, and this was a presentation of my product idea and features that I delivered to MYP Year 2 students in my school. After this presentation (Figure C.2), I gained immense clarity on the various ways through which I could ensure the features of my product would be satisfactory, and most importantly, would perform well against the success criteria. The biggest highlight of this experience was having the opportunity to take the opinions of others, and not just adhering to my own judgement, as well as actively While I did agree with most of the points brought up, I did disagree with a few others, however, at the end of the day, I did have an accepting mindset, because I knew that these individuals had a huge say in the features and idea of the product, as they are target audience it serves. As a result, I developed two vital attributes of the IB Learning profile, becoming a more **Open-Minded** individual by ***seeking and evaluating a range of points of view***, as well as a better **Communicator** by ***collaborating effectively, listening carefully to the perspectives of other individuals and groups***.

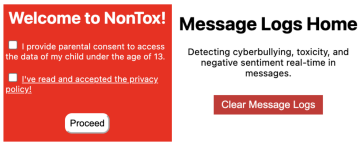


Figure C.3: Final Browser Extension Pages

Next, I was able to become a better **Thinker,** learning to ***use critical and creative thinking skills to analyse and take responsible action on complex problems***, as well as an improved **Inquirer**, wherein I ***nurtured my curiosity, developing skills for inquiry and research.*** These were demonstrated throughout the process of procuring learnings and developing the product, and the best examples to culminate this process are the screenshots of my final browser extension (Figure C.3). From brainstorming unique features, to indulging in active and hands-on product development, the process of developing my product was far from easy, with many problems and objectives requiring novel and well-thought solutions. I was able to demonstrate this in specific instances during the development of my product. When I was deciding the features for my product, I thoroughly and critically analysed the problem of unmonitored and uncontrolled cyberbullying and online toxicity using helpful primary and secondary research and inquiry techniques, to come to the understanding that the lack of real-time response to these issues and limited opportunities for parental involvement in them were the root causes, allowing me to finalize features which included these elements. Most importantly, through exploring these extensive topics of Artificial Intelligence, Machine Learning, Natural Language Processing, and Societal Benefit, I was able to nurture the very curiosity that drove me to initiate this project, leaving me with immense satisfaction of applying and developing my Inquiry and Thinking skills and abilities.

Finally, following my personal project, I have managed my series of academic and personal commitments together in a much more effective manner, primarily due to my increased **Reflective** abilities, being able to ***work to understand my strengths and weaknesses in order to support our learning and personal development***, identifying key areas of both development as well as improvement (Figure C.4). My analysis mainly included the assessment of the various educational abilities that I possessed, and most importantly, the identification of a few crucial soft skills to improve upon, namely managerial skills, overall helping me understand what kind of a student and person I was. At the end of the process, I had been able to make strides to become a more **Balanced** learner, working on developing a better sense of awareness on different techniques of maintaining an equilibrium between the intellectual, physical, and emotional factors of my routine, as well as making significant efforts to practise them in my daily life in the form of calendars and planners, ultimately ***understanding the importance of balancing different aspects of our lives to achieve well-being.***

	Academic	Personal
Strengths	→ Through my personal project, I was able to pursue immense developments in my research and thinking skills. → These were specifically along the lines of using the principles of critical-thinking and problem-solving in order to solve a variety of problems and scenarios. → Furthermore, I was able to explore the interdisciplinary links involved in the various elements of my work. → Finally, I also worked on adapting to various new learning strategies dependent on the different tasks at hand.	→ After interacting with members of my target audience, peers and classmates, subject matter experts, and mentors, I have significantly improved my interpersonal communication. → I have attained the ability of facilitating more engaging and insightful conversations with people of diverse demographics and backgrounds. → As a result, I have also been able to improve my presentation skills, delivering complex ideas in interpretable and concise manners. → Finally, I have become more empathetic to various individuals and societies facing a diverse array of societal issues.
Weaknesses	→ After spending months of time and effort on my personal project, I had come to an understanding that I was mainly limited by the lack of effective time-management and productivity techniques. → While I was able to implement certain frameworks like the Eisenhower matrix, they only helped in prioritizing my tasks, rather than pushing me to complete them, often resulting in me getting distracted or committing to other tasks. → That is why, employing other tools like site-blockers or Pomodoro timers would have allowed the learning process to be more successful.	→ The main area which I have scope to improve in, is being able to ensure that I have greater equipoise between the scholastic and extracurricular aspects of my daily life. → While I did have some planning systems in place, they weren't as comprehensive and applicable as I would have liked them to be. → This was mainly because I had inconsistent levels of focus on different tasks and didn't effectively use time-blocking strategies, often resulting in burnout and stress. → This has made me realize the importance of Meditation and Spirituality, two aspects which weren't part of my project routine. → Retrospectively, managing as well as outlining my preferences beforehand would have allowed me to make more time to work on a variety of obligations, and assisted me in managing my overall well-being.

Figure C.4: Strengths and Weakness Analysis

Overall, like all the activities and tasks I took part in throughout my MYP studies, I also attained and developed numerous attributes of the IB Learner Profile while working on the personal project. In all my endeavors as an MYP student, and especially during the process of my personal project, all of these attributes allowed me to ensure that any task that I undertook, the challenges associated with it, as well as my final solution and output generated to accomplish the goal, would not only result in progress being made in context of the overall personal project, but would also act as a guide, providing me with a sense of motivation as well as a form of understanding on the true purpose and function of the personal project. This ultimately allowed me to make progress in fulfilling the vast components of the rewarding MYP programme, as well as setting a positive framework for achieving success in the forthcoming Diploma Program. For a long time, I always lived by the quote, 'Success has no full stops, it only has commas.', and after the entire process of this personal project, I came to the realization that this journey has served as a major comma in my life, setting me up for future success in the field of Artificial Intelligence Development, one where I hope to explore a career in.

<u>Strand ii. Evaluate the product based on the success criteria</u>

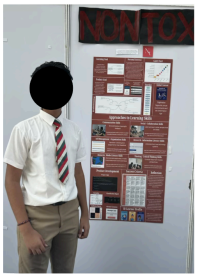
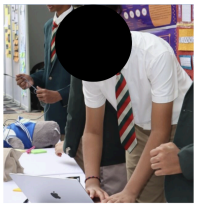


Figure C.5: Personal Project Exhibition Chart

As mentioned earlier, one of the best experiences I had while working on the personal project was my school's annual Personal Project Exhibition (Figure C.5). As a part of the exhibition, I was not only able to comprehend the sheer significance and real-world applicability of my project, but was also able to engage in active interactions, discussions, and conversations (Figure C.6), with members of my target audience as well as parent stakeholders, with all of them allowing me to examine my culminated product from a direct as well as diverse perspectives, to understand where it stands, and how to improve or extend it.


Figure C.6: Engagement in Exhibition

In order to evaluate my final product against my previously-outlined success criteria (Figure A.6), I had identified select testing methods (Figure A.8), for each of my six criteria. I ensured that all of these methods were objective in nature, either by obtaining an appraisal from relevant individuals, such as my target audience, subject-matter specific facilitators, and industry specialists, or by using raw and unfiltered numerical data, ultimately assessing my outcome through methods which were both credible and logical (Figure C.7).

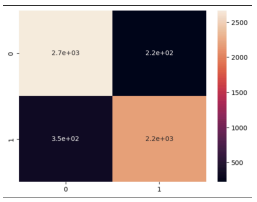
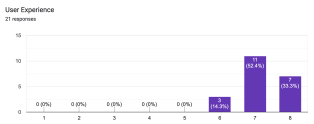
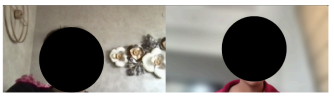
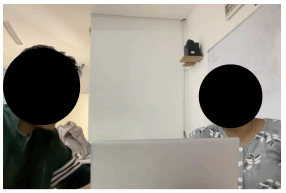
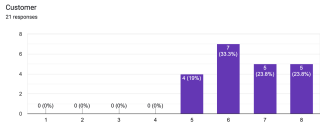
Criterion	Grade	Justification	Improvements and Extensions	Evidence
Model Performance Metrics	8	To reach an achievement level of 7-8, all my values for the model performance metrics, namely Accuracy, Precision, Eecall, and F1 Score, needed to be between 80% and 100%. All my values for these metrics ranged from 86% to 92%, with a vast majority of them concentrated around the 90% mark. Since most of these values were towards the later half of the range, being closer to the goal of 100%, the final grade that I was able to award for this criterion was an 8.	Since I attained an achievement of 8 for the criterion of Model Performance Metrics, my main focus is on extensions. While I have calculated values for accuracy, precision, recall, f1-score, and plotted a confusion matrix, I can further extend this by obtaining values for three more specific metrics, which are AUC-ROC curve, Sensitivity, and Specificity, allowing to get more insights while investigating the models ability to distinguish between positive and negative cases.	 precision recall f1-score 0 0.88 0.92 0.90 1 0.91 0.86 0.88 accuracy 0.89 macro avg 0.90 0.89 0.89 weighted avg 0.89 0.89 0.89
User Experience	7	In order to attain an achievement level of 7-8, my target audience needed to find the overall experience of the application exceptional. After testing the browser extension on their unique cases as well as examining and understanding the functionality of it, the target audience group allowed me to obtain their unfiltered and passionate opinions. The user experience was mainly strengthened due to its ability to perform the tasks of detecting and displaying cyberbullying and toxicity in messages transmitted across these platforms in a variety of applications attempted by the target audience. The only area which was found lacking was related to the responsiveness and dynamics associated with the nature of browser extensions, resulting in a final grade of 7.	While I garnered mostly positive feedback from my target audience, the main aspect of my user experience which they felt could be extended was the overall scope and structure of the extension. This involves creating a mobile app in addition to the browser extension, which will allow for the detection of cyberbullying and toxicity in more applications such as Instagram and Whatsapp, also creating an overall seamless experience for the target audience as well as their parents, with the potential for login and multi-device elements.	
Cost	8	The only requirements for the Cost criterion were that the total monetary expenses spent on procuring the learnings and developing the learnings were lesser than 1000 Indian Rupees. With 838 of them spent on a course on Natural Language Processing for the development of my AI/ML/NLP model, and 150 on a laptop cooling pad to handle the overheating caused by the robust applications, My final price was 988 Indian Rupees, which was indeed under the final budget, resulting in a final grade of 8.	As I was able to keep my cost below the budget which I had outlined, my extensions to this criterion involve working on spending a lesser and saving a greater amount of money form the fixed budget, in order to obtain extra funds which can be set aside to donate to funds or non-profit organisations to assist individuals and societies suffering from a plethora of societal issues, including mental health.	Cost | Price (INR) Natural Language Processing Course | 838 Chrome Extension Development Course | 0 Laptop Cooling Pad | 150 988 | Total Price
Privacy and Safety	7	Achieving a level of 7-8 in this criterion required the presentation of the data privacy and protection information protection to be highly effective. Based on the expert's comments, the privacy policy which I outlined as a part of my browser extension aligned with the principles of the universally-accepted California Consumer Privacy Act and the General Data Protection Regulation, strengthening its overall quality. Furthermore, they stated that by adding a feature of requiring the user to accept the privacy policy and provide parental consent before progressing to using the extension, I was able to ensure that these policies were highlighted and adhered to. However, they also mentioned a minor area which it lacked was the fact that this only represented a privacy policy, while having the potential to represent many others as well at the same time, resulting in a grade of 7.	The main improvements and extensions which I was able to understand were mainly associated with adding a greater amount of protection information policies. This includes Cookies Policy, Terms and Conditions, as well as End-User Licence Agreement to ensure that my browser extension will be able to exhaustively cover a few additional policies.	
Function	7	The specifications involved for achieving a level of 7-8 for Function mainly surrounded the browser extension detecting cyberbullying and displaying the messages in a highly effective manner. After interacting with a Digital Design Facilitator, their expertise brought me to the fact that the workflow of my entire product, involving the browser extensions and AI/ML/NLP model was incredibly effective, efficient, and simple, a factor which strengthened my product the most. The Facilitator spoke about how they were most impressed by the feature of detecting and displaying the messages containing cyberbullying and toxicity in a real-time manner, to make significant strides towards solving a pressing problem. They also stated that working on training my AI/ML/NLP using multiple data modalities would allow to improve the overall functionality, thereby resulting in a grade of 7.	After presenting my final product, I have been able to identify a few improvements and extensions based on the feedback as well as ideas exchanged. These mainly revolve around ensuring that the browser extension doesn't just detect cyberbullying in the text transmitted over chat windows, but also in the verbal speech, using speech-to-text and transcription technologies, as well as hand gestures made, using computer vision.	
Customer	6	To achieve the highest level for this criterion, the browser extension needed to be highly accessible and the consumer group needed to be highly satisfied. However, I fell just short of this band, ending up with the product being adequately accessible and the target audience being moderately satisfied. The main reason which the target audience brought up was the fact that the browser extension would only function on one online communication platform, which was Google Meet, resulting in its applicability and accessibility being reduced to a certain extent. However, a majority of the audience group spoke about how the final features of the product matched the preferences which they expressed earlier, and heavily appreciated the product in that matter, resulting in a final marking of 6.	After my interaction with the target audience, I have come to the understanding that this is the criterion which requires a greater amount of improvement. These include ensuring that the browser extension can access more online communication platforms, such as Zoom, Microsoft Teams, and Slack. Furthermore, I should explore adding password protection to the message logs to ensure that only relevant individuals can access the information, and that it doesn't fall into the wrong hands, to boost satisfaction.	

Figure: C.7: Table Outlining Evaluations, Extensions, and Evidences